

Thuc Nguyen Phong Pham

University of British Columbia | 3rd Year

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GitHub: github.com/prcpham-dev **Portfolio:** prcpham.com

Experience

Software Engineering Intern – Frontend Focus

Apr 2023 – Aug 2023

Tinh Van Software

- Developed responsive web components using React.js, Node.js, TypeScript, Bootstrap, and Tailwind CSS for a client-facing platform.
- Collaborated with backend engineers to integrate and test RESTful APIs using Postman.
- Participated in deployment and planning using Git.
- Gained hands-on exposure to backend infrastructure involving FastAPI and Uvicorn.

Software Engineering Intern – Full-Stack

Apr 2024 – Aug 2024

Tinh Van Software

- Designed and developed a Retrieval-Augmented Generation (RAG) web app with PDF uploader and contextual AI chat capabilities.
- Implemented backend in Python using FastAPI, LangChain, and use Google Gemini to support semantic search and response generation.
- Integrated Redis and Celery for asynchronous task queues, improving responsiveness and scalability.
- Built a responsive frontend using React.js and Bootstrap.
- Deployed on a cloud VPS using Docker, Nginx, HTTPS (SSL), and environment-managed API keys.
- Created RESTful API endpoints for document processing and chat flow, supporting up to 3MB PDF uploads.

Projects

ChatDocument

FastAPI, LangChain, Redis, Celery, React.js, Bootstrap, Docker, Nginx

- Developed a Retrieval-Augmented Generation (RAG) app enabling users to upload and analyze PDFs with contextual AI chat.
- Integrated Google Gemini (2M tokens) with LangChain for embeddings, semantic search, and long-context reasoning.
- Built a scalable backend with FastAPI, offloading heavy tasks asynchronously using Redis and Celery.
- Designed a responsive frontend in React.js and Bootstrap for file upload, chat interaction, and insights display.
- Deployed on a VPS with Docker, Nginx, and SSL certificates, securing API keys via environment management.

InstaSave

Chrome Extension, JavaScript, MutationObserver

- Created a browser extension that injects a “Save” button into Instagram posts and stories for quick media downloads.
- Used DOM manipulation and MutationObserver to dynamically detect and modify Instagram’s interface.
- Delivered lightweight functionality to open or save images and videos directly from the browser.

Shufa Character Downloader

Selenium, asyncio, Tkinter, ChromeDriver

- Built a desktop application with a Tkinter-based UI to automate downloading of Chinese calligraphy images.
- Automated scraping with Selenium WebDriver to fetch and filter results for the selected calligrapher.
- Leveraged asyncio to run multiple download tasks concurrently without blocking the UI.
- Organized images into a structured local directory and displayed progress and logs in the UI.

GhostRun

HTML5 Canvas, JavaScript, LocalStorage

- Built an endless runner game where a ghost character flips gravity to dodge pillars, inspired by Flappy Bird.
- Implemented physics with gravity inversion, collision detection, and difficulty scaling using vanilla JavaScript.
- Designed game loop with `requestAnimationFrame` for smooth rendering on HTML5 Canvas.
- Added HUD with score, high score persistence via `localStorage`, and interactive pause/restart controls.
- Deployed on GitHub Pages for browser play: prcpham-dev.github.io/GhostRun.

SpeedCube

React.js, Three.js, Tailwind CSS

- Built an interactive 3D Rubik's Cube simulator with real-time rendering in a React web app.
- Implemented a precision timer, customizable keybinds, and state control (shuffle, reset).
- Designed a modern, responsive UI with Tailwind CSS for smooth cross-device use.

YoutubeShorts_ClipMixer

OpenAI API, AssemblyAI, Tkinter, MoviePy

- Built a GUI-based tool to automate YouTube Shorts creation from user-provided or AI-generated scripts.
- Integrated APIs for narration, transcription, and captions to streamline video production.
- Assembled and edited clips with MoviePy and FFmpeg, adding background music and rendering final output.
- Provided a Tkinter interface to manage script input, API keys, and video preview/editing.

Education

University of British Columbia

Expected May 2026

Bachelor of Science in Combined major science

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning

Skills

- **Languages:** JavaScript, TypeScript, Python, Java, Swift, SQL, HTML/CSS, C++
- **Frameworks/Libraries:** React.js, Node.js, SwiftUI, Tailwind CSS, Three.js, MoviePy, LangChain, FastAPI, Express.js
- **Tools/Technologies:** Docker, Git, Postman, Redis, Celery, Uvicorn, REST APIs, OpenAI API, AssemblyAI, Selenium/WebDriver, PostgreSQL, AWS